

S3 Cost Deep Dive

AWS Pricing Calculator used for reference.

Scenario testing calculates the cost of storing 1 TB for 12 months.

Scenario A: All S3 Standard

Storage	$1 \text{ TB} \times \$0.023 / \text{GB} =$	\$23.55
PUT requests	$10,000 \times \$0.005 / 1,000 =$	\$0.05
GET requests	$100,000 \times \$0.0004 / 1,000 =$	\$0.04

Monthly total \$23.64
Annual total \$283.68

Scenario B: S3 Standard for month 1,
then S3-IA for months 2-12

Month 1: Standard	1TB x \$0.023	=	\$23.55
Months 2-12: S3-IA	1TB x \$0.0125	=	\$12.80/mo
S3-IA retrieval fee	\$0.01/GB x $\approx 50GB$ retrieved /mo	=	\$0.50/mo
IA GET requests	50,000 x \$0.0001/1000	=	\$0.05/mo

Annual total ~\$170

Scenario C: Standard $\rightarrow$ IA after 30 days
 $\rightarrow$ Glacier Flexible after 90 days

Month 1: Standard	1TB $\times$ \$0.023	=	\$23.55
Months 2-3: S3-IA	1TB $\times$ \$0.0125	=	\$12.80 ea
Months 4-12: Glacier Flex	1TB $\times$ \$0.0036	=	\$3.69 ea
Glacier retrieval 1x/yr (100 GB bulk)	\$0.03/GB $\times$ 100GB	=	\$3.00

Annual total $\sim$ \$85

Scenario D: Intelligent-Tiering (let AWS decide)

Monitoring fee $\sim 1\text{M objects} \times \$0.0025 / 1\text{K tokens} = \$2.56 / \text{mo}$

Storage (auto-tiered) Similar to Scenario B/C = $\sim \$12.35 / \text{mo}$
automatically

Retrieval fees None in I-T = $\$0$

Annual total (est) $\sim \$170 - 200$

Conclusion

Lifecycle approach (Scenario C) saves $\sim 70\%$ versus all-Standard, but requires you to predict access patterns. Intelligent-Tiering is simpler but has monitoring overhead.